
Amazon Kinesis – Detailed Overview

Amazon Kinesis is a fully managed, scalable, and durable service for real-time data streaming on AWS. It allows you to collect, process, and analyze streaming data (logs, events, metrics, video, etc.) as it is generated, enabling immediate insights and responses.

Core Components of Kinesis

Component	Description	Use Case
Kinesis Data Streams (KDS)	Real-time data stream processing with custom consumers (like Lambda, EC2).	Complex processing with low latency
Kinesis Data Firehose (KDF)	Serverless streaming ingestion + automatic delivery to S3, Redshift, etc.	Simplified ETL, no code
Kinesis Data Analytics (KDA)	Run real-time SQL on streaming data from KDS or Firehose.	Real-time dashboards, filtering
Kinesis Video Streams (KVS)	Ingest, store, and analyze video streams from devices.	IoT, security cameras, ML vision

1. Kinesis Data Streams (KDS)

Use Cases:

- Application logs
- Clickstream analytics

- Fraud detection
- IoT sensor streams

Key Concepts:

- **Shard:** A unit of parallelism. Each shard supports 1 MB/s write, 2 MB/s read.
 - **Record:** A unit of data with a partition key, sequence number, and payload.
 - **Retention:** Default 24 hours, can be extended to 7 days.
 - **Consumers:** EC2, Lambda, Kinesis Data Analytics, etc.
 - **Enhanced Fan-Out:** Allows multiple consumers to get data at full speed without shared throughput.
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2. Kinesis Data Firehose (KDF)

Use Cases:

- Serverless ingestion into S3, Redshift, OpenSearch
- Simple log delivery pipelines

Features:

- No need to write producer/consumer code
- Data transformation using Lambda

- Compression (GZIP, Snappy), encryption (KMS), and format conversion (JSON, Parquet, ORC)
- Buffering configuration (1–128 MB or 60–900 seconds)

Supported Destinations:

- Amazon S3
 - Amazon Redshift
 - Amazon OpenSearch Service
 - Generic HTTP Endpoints
 - Splunk
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3. Kinesis Data Analytics (KDA)

Use Cases:

- Real-time dashboards
- Streaming data aggregations
- Event filtering or alerting

Features:

- Write SQL-like queries to analyze data from KDS/KDF

- Use sliding windows, tumbling windows for time-series processing
 - Integrated with AWS CloudWatch for monitoring
 - Output to Firehose, Lambda, S3, or Kinesis Streams
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4. Kinesis Video Streams (KVS)

Use Cases:

- Video monitoring
- Live stream ingestion for ML
- Video archiving

Features:

- Streams video, audio, and time-encoded data from edge devices
 - Integrates with Amazon Rekognition, SageMaker
 - HLS playback support
 - Real-time or on-demand video analytics
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Security & Compliance

-  **Encryption at Rest:** AWS KMS (default enabled for Firehose)
 -  **Encryption in Transit:** TLS
 -  **VPC Endpoints:** Private connectivity
 -  **IAM Policies:** Fine-grained access control
 -  **CloudTrail & CloudWatch Logs:** For auditing and monitoring
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Kinesis vs Kafka vs Firehose

Feature	Kinesis Streams	Kafka	Kinesis Firehose
Management	Fully managed	Self-managed or MSK	Fully managed
Code Required	Yes	Yes	No
Destination	Custom apps	Custom apps	S3, Redshift, OpenSearch
Replay Support	Yes (24h–7d)	Yes	No
Processing	Real-time	Real-time	Near real-time
Use Case	Custom ingestion + processing	Large scale streaming platforms	Easy delivery pipelines

Pricing Highlights

- **Data Streams:**

- \$0.015 per shard per hour
 - \$0.014 per million PUT payload units
 - **Firehose:**
 - \$0.029 per GB ingested
 - Additional for format conversion & transformation
 - **Analytics:**
 - Based on processing unit hours
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✓ Common Architecture Patterns

1. Real-Time Log Processing

- Web App → Kinesis Streams → Lambda → S3/DynamoDB

2. Serverless ETL

- IoT Devices → Firehose → S3 → Athena/Redshift

3. Streaming Analytics

- Logs → Kinesis Stream → Data Analytics → Redshift Dashboard
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🧠 Best Practices

- Enable **Enhanced Fan-Out** for high consumer throughput
 - Use **partition keys** that evenly distribute data
 - Tune **shard count** based on throughput
 - Use **buffer settings** wisely for Firehose delivery latency
 - Monitor with **CloudWatch metrics**
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